Zhao **Du**

Ph.D. Applicant in Biomedical Engineering, Robotics, and Human-Machine Interfaces

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Education

University of Chinese Academy of Sciences; Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences *Shenzhen, China*

M.Eng. in Control Engineering; GPA: 3.6/4.0 *Sep. 2024 - Expected Jun. 2027*

- Graduate training at SIAT, a Chinese Academy of Sciences research institute with interdisciplinary programs in engineering, biomedical sensing, and intelligent systems; UCAS ranked No. 54 globally in the 2025-2026 U.S. News & World Report Best Global Universities ranking.

Beijing Jiaotong University, School of Electronic and Information Engineering *Beijing, China*

B.Eng. in Automation; GPA: 3.2/4.0 *Sep. 2019 - Jun. 2023*

- Completed engineering coursework in control, signal processing, embedded systems, sensors, algorithms, machine learning, and optimization.
- Beijing Jiaotong University is a national Double First-Class university with engineering strengths in transportation, automation, and information systems.

Publications

Democratizing Precision: High-Performance Hand Gesture Recognition via Low-Cost Exoskeleton and Physics-Informed Graph Neural Networks *IEEE Transactions on Instrumentation and Measurement*

First author; Published; IF 7.0, JCR Q1

- Introduced a low-cost hand-motion sensing pipeline with a 3-D-printed exoskeleton and a physics-informed graph network that encodes hand-joint structure and biomechanical constraints.
- Demonstrated strong cross-user generalization and data efficiency, supporting affordable gesture-recognition systems for wearable interfaces, rehabilitation training, and assistive devices.

Cross-Subject Hand Gesture Recognition via Inter-Modal Alignment of sEMG and A-mode Ultrasound for Subject-Invariant Robotic Control *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*

First author; Accepted

- Proposed an inter-modal alignment framework that learns subject-invariant representations from sEMG and A-mode ultrasound, addressing user-to-user signal shift in multimodal physiological sensing.
- Validated the method in offline evaluation and robotic-control experiments, showing practical value for calibration-light assistive human-machine interaction.

Online Hand Gesture Recognition with Hybrid sEMG-AUS Sensing and a Spiking Neural Network for Robotic Applications *IEEE Robotics and Automation Letters*

First author; Invited for second-round review; IF 5.3, JCR Q1 (Robotics)

- Developed a dual-branch spiking neural network that jointly processes electrophysiological and morphological features from hybrid sEMG-AUS sensing for online hand gesture recognition.
- Targets low-latency and energy-efficient embedded deployment, providing a practical path toward wearable prosthetic interfaces and robotic-hand control.

Experience

CRRC Zhuzhou Locomotive Co Ltd, Urban Rail Development Department *Zhuzhou, China*

Engineer; Urban Rail Development and Engineering Documentation *Aug. 2023 - Jun. 2024*

- Tasked with supporting technical documentation, interface coordination, and engineering drawing review for urban rail vehicle development.
- Prepared routine engineering records and review materials, coordinating feedback among mechanical, electrical, and systems engineering teams.
- Served as a documentation and workflow-oriented engineer, using CAD/UG tools and spreadsheet automation to organize design files, compare versions, and improve recurring review workflows.

NIO, Assisted and Intelligent Driving Division *Beijing, China*

Data Engineering Intern; Data Pipeline and Tooling Contributor *Mar. 2025 - Jul. 2025*

- Tasked with supporting large-scale engineering data analysis, quality checking, and visualization for intelligent-driving development workflows.
- Built Python/Linux tools for data retrieval, statistics, configuration updates, and result visualization to improve routine data-inspection efficiency.
- Worked as a tooling-oriented engineering contributor, connecting data requirements with practical scripts and reproducible analysis workflows.

Leadership

School of Artificial Intelligence, University of Chinese Academy of Sciences *Beijing, China*

President of the Graduate Student Union *2024 - 2025*

- Led a 30-member graduate student union serving approximately 500 students, coordinating the publicity, sports and arts, and academic departments.
- Organized the “Artificial Intelligence+” academic salon series, coordinating speakers, promotion, venues, and student participation for interdisciplinary discussion among graduate students.
- Coordinated a cross-school evening gala with four schools, approximately 1,000 audience members, and about 150 student performers, managing program review, rehearsal scheduling, and on-site execution.
- Planned school-level sports and community activities, including badminton selection and inter-team matches, to support student engagement and cross-cohort communication.

Awards

2025	**National Scholarship**, Ministry of Education of China	*China*
2025	**Outstanding Student**, University of Chinese Academy of Sciences	*China*
2025	**Outstanding Student Leader**, University of Chinese Academy of Sciences; selected among the top 5% of student leaders	*China*
2022	**Second Prize**, Lanqiao Cup Algorithm Competition, Beijing Division	*China*

Skills

Programming and Robotics	Python, ROS, C/C++
Learning Models	Graph networks, spiking neural networks